CoveredNow

Implementation Plan - a FirmwareNerds product. Zimbabwe-only launch for university, college and high school fees. Prepared for Biggie, November 2026.

1. Vision and scope

One-line pitch: verified students (or their guardians) list school-confirmed fee bills, and donors, local and in the diaspora, pay the school directly and climb a leaderboard as they do it.

Core principles

- **Pay the institution, never the student.** Money flows donor to platform payment partner to school account.
- **The school confirms every bill.** The confirmed amount is the only amount a donor sees.
- **100% of the confirmed fee reaches the school.** Revenue comes from an optional donor-paid fee, sponsors and school services, never from the bill itself.
- **Dignity first.** Students are never ranked, shamed or publicly identified by default.
- **Start narrow.** A handful of institutions, manual operations where automation is not yet proven.

Out of scope for v1: non-fee bills (rent, medical, utilities), cash payouts to individuals, a native mobile app, multi-country support.

2. Users and roles

Role	Who	What they do
Student / Guardian	University and college students; guardians of high school students	Register, verify identity, submit a fee bill, post updates, thank donors
Donor	Local individuals, diaspora, companies, churches, alumni groups	Browse bills, fund fully or partially, set visibility, view impact and rank
School contact	Bursar or accounts officer	Confirm, correct or reject bills; confirm receipt of payment
Admin / Ops	Your team	Onboard schools, review verifications, handle payouts, disputes, moderation
Sponsor	Companies running CSR campaigns	Fund challenges, view impact reports

3. Core user journeys

A. Student submits a bill

1. Sign up with phone number (OTP) and email.
2. Complete identity verification (national ID or birth certificate, student ID). High school students: the guardian owns the account and uploads their own ID plus the student's details.
3. Pick the institution from the verified list, enter reg number, term, amount and upload the fee statement.
4. Bill enters status PENDING_SCHOOL.

B. School confirms

1. A verified school contact receives a WhatsApp or email message with a signed, expiring link.
2. The page shows student name, reg number and claimed amount. Buttons: Confirm, Wrong amount (enter the correct figure), Not our student.

3. The result is stored with who, when, amount and term. The bill moves to SCHOOL_CONFIRMED and is published to the feed.

C. Donor funds a bill

1. Browse the feed (filter by level, institution, amount, urgency).
2. Choose an amount (full or partial), choose visibility (public name, alias, anonymous), optionally add the platform tip.
3. Pay via Paynow (EcoCash, OneMoney, cards) or card for diaspora donors.
4. Funds are recorded against the bill. When fully funded, the bill moves to FUNDED and closes to new pledges.

D. Payout and proof

1. Ops (manual at first, automated later) pays the school's registered account.
2. Ops uploads proof of payment; the bill becomes PAID.
3. The school confirms receipt; the bill becomes RECEIPT_CONFIRMED. Only now does the donation count toward leaderboards and the donor sees the receipt.

E. Close the loop

The student posts a short, moderated update or thank-you. Donors are notified. This is the main reason to return.

4. Bill status lifecycle

```
DRAFT > PENDING_SCHOOL > SCHOOL_CONFIRMED > PARTIALLY_FUNDED > FUNDED > PAID > RECEIPT_CONFIRMED
```

Side exits: REJECTED (school or admin), EXPIRED (past due with no funding, can be relisted), CANCELLED (student withdraws), DISPUTED (payment or amount problem).

5. Architecture and stack

Chosen to match your existing strengths so you move fast.

- **Frontend:** Next.js (App Router) + TypeScript + Tailwind. Lucide icons throughout the UI, no emojis.
- **Backend:** Next.js route handlers and server actions for the MVP; split a Node.js worker service for jobs once volume grows.
- **Database:** PostgreSQL with Prisma.
- **Auth:** Clerk or NextAuth with phone OTP as the primary method (most users are mobile-first).
- **File storage:** S3-compatible object storage with private buckets and signed URLs for IDs and fee statements.
- **Jobs and queues:** BullMQ + Redis (or a hosted alternative) for notifications, leaderboard snapshots, expiry and reconciliation.
- **Messaging:** WhatsApp Business API (via a provider) as the main channel to schools and donors; email and SMS as fallback.
- **Payments:** Paynow for local collection; a card processor for diaspora payments if available to your entity (see Section 8).
- **Observability:** Sentry, structured logs, uptime monitoring, an audit-log table for every sensitive action.
- **Hosting:** Vercel or a VPS for the app, managed Postgres, nightly backups with restore tested before launch.

Performance note: design for low bandwidth. Server-render the feed, compress images, keep the first load light, and make every key flow work well on a mid-range Android phone over 3G/4G.

6. Data model (Prisma sketch)

```
enum Role { STUDENT GUARDIAN DONOR SCHOOL_CONTACT ADMIN SPONSOR }
enum BillStatus { DRAFT PENDING_SCHOOL SCHOOL_CONFIRMED PARTIALLY_FUNDED
                  FUNDED PAID RECEIPT_CONFIRMED REJECTED EXPIRED CANCELLED DISPUTED }
enum Visibility { PUBLIC ALIAS ANONYMOUS }

model User {
  id String @id @default(cuid())
  role Role
  phone String @unique
  email String?
  name String
  createdAt DateTime @default(now())
  verification Verification?
  donorProfile DonorProfile?
}

model Verification {
  id String @id @default(cuid())
  userId String @unique
  user User @relation(fields: [userId], references: [id])
  status String            // PENDING | APPROVED | REJECTED
  idDocUrl String
  reviewedBy String?
  reviewedAt DateTime?
}

model Biller {
  id String @id @default(cuid())
  name String
  type String              // UNIVERSITY | COLLEGE | HIGH_SCHOOL (more later)
  city String
  verified Boolean @default(false)
  payoutMethod String      // BANK | MOBILE_MONEY | MERCHANT
  payoutDetails Json       // encrypted at rest, set by ops only
  contacts BillerContact[]
  bills Bill[]
}

model BillerContact {
  id String @id @default(cuid())
  billerId String
  biller Biller @relation(fields: [billerId], references: [id])
  name String
  phone String
  email String?
  roleTitle String
  verifiedAt DateTime?
}

model Bill {
  id String @id @default(cuid())
  ownerId String
  billerId String
  biller Biller @relation(fields: [billerId], references: [id])
  regNumber String
  term String
  claimedAmount Decimal
  confirmedAmount Decimal?
  currency String @default("USD")
  category String @default("SCHOOL_FEES")
  dueDate DateTime
  documentUrl String
  documentHash String
  status BillStatus @default(DRAFT)
  confirmations BillConfirmation[]
  donations Donation[]
  updates StudentUpdate[]
  createdAt DateTime @default(now())
  @@unique([billerId, regNumber, term])
}

model BillConfirmation {
  id String @id @default(cuid())
```

```
  billId String
  bill Bill @relation(fields: [billId], references: [id])
  contactId String
  status String            // CONFIRMED | CORRECTED | REJECTED | RECEIPT
  confirmedAmount Decimal?
  token String @unique
  expiresAt DateTime
  respondedAt DateTime?
}

model Donation {
  id String @id @default(cuid())
  billId String
  bill Bill @relation(fields: [billId], references: [id])
  donorId String
  amount Decimal
  platformTip Decimal @default(0)
  gatewayFee Decimal @default(0)
  currency String
  visibility Visibility @default(PUBLIC)
  gatewayRef String @unique
  status String            // INITIATED | SETTLED | FAILED | REFUNDED
  settledAt DateTime?
}

model Payout {
  id String @id @default(cuid())
  billId String @unique
  amount Decimal
  method String
  reference String
  proofUrl String
  paidAt DateTime
}

model DonorProfile {
  id String @id @default(cuid())
  userId String @unique
  user User @relation(fields: [userId], references: [id])
  displayName String
  visibility Visibility @default(PUBLIC)
  tier String @default("BRONZE")
  streakWeeks Int @default(0)
  badges DonorBadge[]
}

model Badge { id String @id @default(cuid())
              key String @unique
              name String
              icon String }
model DonorBadge { id String @id @default(cuid())
                   donorId String
                   badgeId String
                   awardedAt DateTime @default(now()) }

model LeaderboardSnapshot {
  id String @id @default(cuid())
  period String            // e.g. 2026-11, 2026-T3, ALL_TIME
  metric String            // TOTAL_GIVEN | BILLS_CLEARED | STREAK | ...
  donorId String
  rank Int
  value Decimal
  @@index([period, metric, rank])
}

model StudentUpdate {
  id String @id @default(cuid())
  billId String
  bill Bill @relation(fields: [billId], references: [id])
  body String
  status String            // PENDING | APPROVED | REJECTED
  createdAt DateTime @default(now())
}

model AuditLog {
  id String @id @default(cuid())
  actorId String?
  action String
  entity String
  entityId String
```

```
  meta Json?
  createdAt DateTime @default(now())
}
```

Key constraints: one bill per biller + reg number + term; payout details are only ever editable by ops; leaderboards count only donations whose bill reached RECEIPT_CONFIRMED.

Why Biller, not Institution: the thing being paid is a biller. Today every biller is a school, but the model and the `category` field are ready for rent, utilities and clinics later. The product, copy and feed stay school-fees-only until the pilot proves the confirmation model.

7. Verification and trust design

Three layers

1. **Institution whitelist.** Only schools you have onboarded and verified can appear. Verify the bursar by phone and against the school's official website or letterhead.
2. **Person verification.** National ID or birth certificate plus student ID or enrollment letter. Manual review queue in the admin dashboard for the MVP; consider a KYC provider once volume justifies it.
3. **School confirmation of each bill.** See Journey B. The school's contact list and payout details live only on your side, so a student can never redirect money.

Fraud controls

- Hash every uploaded document and reject duplicates.
- Unique constraint on institution + reg number + term.
- New-account limits, such as one open bill per student per term.
- Alert on unusual patterns: the same device or phone across many accounts, donor and student sharing a device or payment source.
- Refund or roll over any surplus if a fee changes after funding.

Minors. High school accounts are owned by a guardian. Public listings show first name or initials, school, form or level and amount only: no photo, no surname by default.

8. Payments and payouts

Collection

- Local donors: Paynow (EcoCash, OneMoney, bank cards) as the main rail.
- Diaspora donors: a card processor that works with your legal entity; show the amount in USD and the ZiG equivalent.
- Record every payment with a gateway reference. Reconcile daily against gateway reports.

Holding funds. Prefer a licensed payment partner to hold funds until payout rather than holding money in your own account. Confirm the structure with a lawyer and your payment provider before launch (see Section 11).

Payout (phased)

- MVP: ops pays the school manually (bank transfer, mobile money, merchant code) once a bill is FUNDED, then uploads proof.
- Later: scheduled batch payouts, then API-driven payouts for schools that support them.
- Every payout is recorded with reference, amount and proof, and requires a second admin approval above a set threshold.

Fee math (default). Bill $200: the school receives $200. The donor pays $200 plus gateway cost plus an optional platform tip (suggested around 7%, pre-filled but removable). Show the full breakdown before the donor confirms.

9. Engagement and ranking features

Leaderboards (computed by a scheduled job and cached)

- Total given, bills cleared, current streak, students supported, rising donors.
- Monthly and per-term seasons plus an all-time hall of fame.
- Visibility controls: public name, alias or anonymous (anonymous donors still see their private rank).
- Rank credit only after the school confirms receipt.

Daily-return hooks

- Impact updates from students and schools (moderated).
- Bill of the Day, almost-funded bills, real due-date countdowns.
- Match challenges and weekly community goals.
- Streaks with a freeze option, tiers and badges (Lucide icons).
- Team boards for companies, churches, families and alumni groups.
- Shareable WhatsApp cards ("I helped clear a term of fees").
- Seasonal pushes around term openings and exam-fee deadlines.

Guardrails

- No fake urgency or guilt copy; countdowns reflect real dates only.
- Donor-set monthly caps and easy recurring giving.
- Students are never ranked or named in connection with who paid.

10. Monetization plan

Stage	Revenue	Notes
MVP	Optional donor-paid platform tip (default around 7%)	Transparent breakdown; pass gateway cost to the donor
Growth	Corporate sponsorship of challenges and team boards	CSR budgets from banks, telecoms, insurers, retail
Growth	Patron tier (analytics, statements, branded reports)	Subscription for donors and companies
Scale	School services (auto balance confirmation, reconciliation, fee-system integration)	Free basic confirmation for all schools

Avoid: ads, cutting into the bill, selling data, and (until legally cleared) float or exchange-rate income.

11. Legal, compliance and governance

Talk to a Zimbabwean lawyer and accountant **before** taking real money. Questions to settle:

1. **Entity structure.** A for-profit company charging service fees versus a registered trust or PVO receiving donations; many platforms use a non-profit receiver plus a technology company.
2. **Payments regulation.** Whether holding and forwarding funds needs Reserve Bank of Zimbabwe approval or a licensed partner; handling of USD from the diaspora and exchange-control rules.
3. **Tax.** ZIMRA treatment of donations, platform fees and payouts to schools; whether donors can get receipts for tax purposes.

4. **Data protection.** Compliance with the Cyber and Data Protection Act: lawful basis, consent, retention limits, data subject rights, and a named data protection contact.
5. **Child safeguarding.** Policies for minors' data, guardian consent, moderation of updates and a reporting process.
6. **Terms and policies.** Terms of service, privacy policy, refund and dispute policy, school agreement, and donor disclosure that donations are gifts paid to the institution.

12. Security and privacy

- Encrypt IDs, documents and payout details at rest; private storage buckets with short-lived signed URLs.
- Role-based access; admins see only what they need; every sensitive read and write goes to the audit log.
- Signed, single-use, expiring tokens for school confirmation links; rate-limit and log all attempts.
- Rate limiting, bot protection and OTP abuse controls on sign-up.
- Never log full ID numbers or document contents. Define retention (for example, delete rejected-verification documents after a set period).
- Back up daily; test a restore before launch; document an incident-response plan.
- Webhook signature verification and idempotency keys on all payment callbacks.

13. Phased roadmap

Timelines assume one full-time developer (you) plus part-time help for ops and design; adjust as needed.

Phase 0: Validate and lay groundwork (weeks 0 to 4)

- Meet 5 to 8 bursars across at least one university, one college and two high schools; confirm they will confirm bills and receive payments from a third party.
- Interview 20 prospective donors (local and diaspora) and 15 students or guardians.
- Choose a payment partner and confirm how funds can be held and paid out.
- Engage a lawyer and accountant; decide entity structure.
- Produce clickable designs for the five core flows.
- **Exit criteria:** at least 3 institutions committed, a payments route confirmed, and a legal path agreed.

Phase 1: MVP build (weeks 5 to 12)

- Auth with phone OTP and role handling.
- Student and guardian onboarding and verification upload.
- Institution and contact management (admin).
- Bill submission and school confirmation flow (WhatsApp and email links).
- Public feed with filters; bill detail page; partial funding.
- Paynow checkout, webhooks, donation ledger.
- Admin dashboard: verification queue, bill review, payout recording with proof, dispute handling.
- Receipts and basic notifications.
- **Exit criteria:** a full end-to-end test (bill, confirmation, donation, payout, receipt confirmation) with real small amounts and a friendly school.

Phase 2: Closed pilot (weeks 13 to 18)

- 3 to 5 institutions, 30 to 50 bills, invited donors only.
- Daily reconciliation, manual payouts, weekly review of every dispute and delay.
- Add: impact updates with moderation, simple donor dashboard, first leaderboards (total given and bills cleared).
- **Exit criteria:** at least 25 bills fully paid and confirmed by schools, zero unresolved fraud cases, payout time under 3 working days, donor feedback collected.

Phase 3: Public launch (weeks 19 to 28)

- Open donor sign-up; launch before a term opening to capture peak need.
- Add: all leaderboards, tiers and badges, streaks, Bill of the Day, almost-funded bills, WhatsApp share cards, diaspora card payments, recurring giving.
- Start corporate sponsor outreach with pilot results as proof.
- Automate notifications and leaderboard snapshots; add monitoring and alerts.
- **Exit criteria:** repeat donors above target, payout automation for batch payments, stable operations without daily founder intervention.

Phase 4: Scale (months 8 to 18)

- Onboard more institutions per city, then per province.
- School-side tools: balance confirmation integration, reconciliation reports (paid add-ons).
- Patron tier and corporate dashboards; team boards and alumni campaigns.
- Consider a mobile or PWA experience, KYC provider integration, USSD or low-data access for students.
- Evaluate expansion to non-fee education costs (uniforms, books, exam fees) once the core is proven.

14. Team and operating cost (indicative)

Roles by phase

- Phase 0 to 1: you (product + engineering), a part-time designer, a part-time ops and support lead, legal and accounting advisers.
- Phase 2 to 3: add a school relationship manager, a support and moderation person, and a part-time second developer.
- Phase 4: partnerships lead for corporate sponsors, finance and reconciliation officer, data protection officer.

Cost categories to budget (get real quotes for each)

- Hosting, database, storage, monitoring, messaging (WhatsApp and SMS) costs
- Payment gateway and bank or mobile-money fees
- Legal registration, advisory and compliance
- Design, moderation and support labour
- Marketing and school or donor events
- A reserve for refunds, disputes and fraud losses

15. Metrics to track

Trust and operations

- Time from submission to school confirmation (target: under 48 hours)
- Time from fully funded to paid (target: under 3 working days)
- Share of bills confirmed by school; rejection and fraud rates
- Dispute and refund rate

Growth and engagement

- Bills listed, funded and cleared per week; funding rate and time to full funding
- Donor conversion (visitor to first donation), repeat-donor rate, donors with an active streak
- Daily and weekly active donors; notification-to-open rate
- Average donation, tip opt-in rate, average tip

Impact and finance

- Total fees cleared, students supported, institutions active
- Revenue per cleared bill versus cost per cleared bill
- Sponsor revenue and Patron subscribers

16. Risks and mitigations

Risk	Likelihood	Mitigation
Schools slow or unwilling to confirm	High	Make it one tap on WhatsApp; pitch it as free fee recovery; onboard bursars personally; offer fee-system integration
Fake or inflated bills	Medium	School confirmation, document hashing, one bill per student per term, ops review, payouts only to school-owned accounts
Donors do not trust a new platform	High	Proof of payment on every bill, school receipt confirmation, pilot with known donors, public transparency report
Regulatory or banking problems	Medium	Use a licensed payment partner; lawyer review before launch; keep payout operations documented
Exchange-rate and currency issues (USD vs ZiG)	Medium	Hold amounts in the currency the school bills in; show equivalents; avoid taking spread income
Child privacy harm	Low but severe	Guardian-owned accounts, minimal public data, moderation, safeguarding policy
Leaderboards cause gaming or pressure	Medium	Rank only on school-confirmed receipts; multiple boards; donor caps; no guilt copy
Unit economics too thin	Medium	Optional tip plus sponsors and Patron tier; keep ops lean and manual until volume justifies automation
Founder bottleneck	High	Admin tooling from day one; document processes; hire ops help in Phase 2

17. Launch readiness checklist

- [ ] Legal entity and payment structure confirmed in writing
- [ ] Terms, privacy policy, school agreement and refund policy published
- [ ] 3+ institutions signed with verified contacts and payout accounts

- [ ] End-to-end test completed with real money
- [ ] Paynow webhooks, idempotency and daily reconciliation working
- [ ] Audit log, backups and restore test done
- [ ] Admin runbooks: verification, payout, dispute, fraud, incident
- [ ] Moderation process for student updates
- [ ] Monitoring and alerts for payment failures and stuck bills
- [ ] Donor support channel (WhatsApp number and email) staffed
- [ ] Launch timed before a term opening

18. Your next 14 days

1. Book meetings with 5 to 8 bursars; bring a one-page explanation and a mock confirmation page on your phone.
2. Speak to Paynow (and any alternatives) about collection, holding funds and payouts.
3. Book a first session with a lawyer on entity structure and payments regulation.
4. Interview 10 potential donors, including at least 3 in the diaspora, about what would make them pay a stranger's fees.
5. Draft wireframes for the five core flows and test them with 5 students or guardians.
6. Set up the repository: Next.js + TypeScript + Prisma + Postgres, the schema from Section 6, auth with phone OTP and a skeleton admin area.
7. Decide a working name, and register the domain and social handles.

This plan is a starting framework. Legal, tax, regulatory and cost figures need confirming with qualified professionals and real quotes before you commit money or collect payments.